ANSWER KEY & MARKING SCHEME SET 1

Chapter 2: **Power Play** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 8	$\frac{1}{a^{-n}} = a^n = 8$
2	**(a)** 81	Even power gives a positive value; odd power gives a negative value
3	**(b)** 6×10^7	Place the decimal point after the first non-zero digit
4	**(d)** 0.00407	Move the decimal point according to the sign of the exponent
5	**(a)** 2^{-6}	$\frac{1}{2^6} = 2^{-6}$
6	**(d)** 7	The number being multiplied repeatedly is the base
7	**(d)** 5	One lakh is 10^5
8	**(b)** 9	$\frac{1}{a^{-n}} = a^n = 9$
9	**(c)** -32	Even power gives a positive value; odd power gives a negative value
10	**(a)** 3.844×10^5	Place the decimal point after the first non-zero digit

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$(2^3)^4 = 2^{12}$ (1); $2^{12} \times 2^{-5} = 2^7$ (1).	1+1
12	$\left(\frac{3}{2}\right)^3$ (1) $= \frac{27}{8}$ (1).	1+1
13	1.496×10^8 (2).	2
14	$(3^2)^5 = 3^{10}$ (1); $3^{10} \times 3^{-4} = 3^6$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	(i) 2^3 (1); (ii) 2^{-6} (1); (iii) 2^7 (1).	1+1+1
16	Add the exponents: $\left(\frac{2}{3}\right)^{-2}$ (1); $= \left(\frac{3}{2}\right)^2$ (1); value $\frac{9}{4}$ (1).	1+1+1
17	(i) 5.9×10^6 (1); (ii) 7.4×10^{-4} (1); (iii) 8×10^4 (1).	1+1+1
	OR: (i) 3^{-2} (1); (ii) 3^{-6} (1); (iii) 3^7 (1).	1+1+1
18	Add the exponents: $\left(\frac{3}{2}\right)^{-3}$ (1); $= \left(\frac{2}{3}\right)^3$ (1); value $\frac{8}{27}$ (1).	1+1+1
	OR: (i) 1.23×10^7 (1); (ii) 5.8×10^{-6} (1); (iii) 6.5×10^3 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $x + 3 = 11$, $x = 8$ (2); (ii) $4x = 20$, $x = 5$ (2); (iii) $x - 2 = -3$, $x = -1$ (1).	2+2+1
	OR: $600000 = 6 \times 10^5$ (1); $2000 = 2 \times 10^3$ (1); product 1.2×10^9 (2); quotient 3×10^2 (1).	1+1+2+1
20	(i) $x + 2 = 9$, $x = 7$ (2); (ii) $5x = 35$, $x = 7$ (2); (iii) $x - 4 = -1$, $x = 3$ (1).	2+2+1
	OR: $80000000 = 8 \times 10^7$ (1); $400 = 4 \times 10^2$ (1); product 3.2×10^{10} (2); quotient 2×10^5 (1).	1+1+2+1

Mistakes to avoid:

- Writing $2^{-3} = -8$ or $-\frac{1}{8}$; a negative exponent means the **reciprocal**, so $2^{-3} = \frac{1}{8}$.
- Adding the bases: $2^3 \times 2^4 = 4^7$ instead of $2^{3+4} = 2^7$.
- Saying $a^0 = 0$; for $a \neq 0$ it is 1.
- Multiplying exponents when multiplying powers of the same base ($a^m \times a^n = a^{mn}$) or adding them in $(a^m)^n$.
- Leaving the answer as 25×10^4 or 0.25×10^6; in standard form the number part must satisfy $1 \leq m < 10$.
- Getting the sign of the exponent wrong for small numbers: $0.0003 = 3 \times 10^{-4}$, not 3×10^4.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME — SET 2

Chapter 2: **Power Play** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 56200	Move the decimal point according to the sign of the exponent
2	**(c)** 3^{-4}	$\frac{1}{3^4} = 3^{-4}$
3	**(b)** 3	The number being multiplied repeatedly is the base
4	**(d)** 7	One crore is 10^7
5	**(d)** 25	$\frac{1}{a^{-n}} = a^n = 25$
6	**(b)** -125	Even power gives a positive value; odd power gives a negative value
7	**(b)** 5.2×10^{-6}	Place the decimal point after the first non-zero digit
8	**(a)** 0.000091	Move the decimal point according to the sign of the exponent
9	**(c)** 5^{-3}	$\frac{1}{5^3} = 5^{-3}$
10	**(c)** 11	The number being multiplied repeatedly is the base

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\left(\frac{2}{5}\right)^2$ (1) $= \frac{4}{25}$ (1).	1+1
12	7.53×10^{-7} (2).	2
13	$(5^2)^3 = 5^6$ (1); $5^6 \times 5^{-2} = 5^4$ (1).	1+1
14	$\left(\frac{3}{4}\right)^2$ (1) $= \frac{9}{16}$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	(i) 5^{-2} (1); (ii) 5^{-3} (1); (iii) 5^4 (1).	1+1+1
16	Add the exponents: $\left(\frac{5}{4}\right)^{-3}$ (1); $= \left(\frac{4}{5}\right)^3$ (1); value $\frac{64}{125}$ (1).	1+1+1
17	(i) 7.5×10^8 (1); (ii) 2.9×10^{-3} (1); (iii) 4.1×10^4 (1).	1+1+1
	OR: (i) 7^2 (1); (ii) 7^{-4} (1); (iii) 7^4 (1).	1+1+1
18	Add the exponents: $\left(\frac{2}{5}\right)^{-3}$ (1); $= \left(\frac{5}{2}\right)^3$ (1); value $\frac{125}{8}$ (1).	1+1+1
	OR: (i) 3.6×10^6 (1); (ii) 4.1×10^{-5} (1); (iii) 9.9×10^5 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $x + 4 = 10$, $x = 6$ (2); (ii) $3x = 18$, $x = 6$ (2); (iii) $x - 3 = -2$, $x = 1$ (1).	2+2+1
	OR: $0.009 = 9 \times 10^{-3}$ (1); $0.00003 = 3 \times 10^{-5}$ (1); product 2.7×10^{-7} (2); quotient 3×10^2 (1).	1+1+2+1
20	(i) $x + 1 = 6$, $x = 5$ (2); (ii) $2x = 14$, $x = 7$ (2); (iii) $x - 5 = -3$, $x = 2$ (1).	2+2+1
	OR: $4500000 = 4.5 \times 10^6$ (1); $150 = 1.5 \times 10^2$ (1); product 6.75×10^8 (2); quotient 3×10^4 (1).	1+1+2+1

Mistakes to avoid:

- Writing $2^{-3} = -8$ or $-\frac{1}{8}$; a negative exponent means the **reciprocal**, so $2^{-3} = \frac{1}{8}$.
- Adding the bases: $2^3 \times 2^4 = 4^7$ instead of $2^{3+4} = 2^7$.
- Saying $a^0 = 0$; for $a \neq 0$ it is 1.
- Multiplying exponents when multiplying powers of the same base ($a^m \times a^n = a^{mn}$) or adding them in $(a^m)^n$.
- Leaving the answer as 25×10^4 or 0.25×10^6; in standard form the number part must satisfy $1 \leq m < 10$.
- Getting the sign of the exponent wrong for small numbers: $0.0003 = 3 \times 10^{-4}$, not 3×10^4.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 2: **Power Play** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 6	One million is 10^6
2	**(b)** 1000	$\frac{1}{a^{-n}} = a^n = 1000$
3	**(a)** 16	Even power gives a positive value; odd power gives a negative value
4	**(b)** 1.496×10^8	Place the decimal point after the first non-zero digit
5	**(b)** 1250000	Move the decimal point according to the sign of the exponent
6	**(b)** 10^{-5}	$\frac{1}{10^5} = 10^{-5}$
7	**(a)** 9	The exponent tells how many times the base is used as a factor
8	**(c)** 9	One billion is 10^9
9	**(d)** 16	$\frac{1}{a^{-n}} = a^n = 16$
10	**(b)** 49	Even power gives a positive value; odd power gives a negative value

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	1.4×10^9 (2).	2
12	$(7^3)^2 = 7^6$ (1); $7^6 \times 7^{-4} = 7^2$ (1).	1+1
13	$\left(\frac{10}{3}\right)^2$ (1) $= \frac{100}{9}$ (1).	1+1
14	7.5×10^{-6} (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	(i) 10^{-3} (1); (ii) 10^{-8} (1); (iii) 10^8 (1).	1+1+1
16	Add the exponents: $\left(\frac{4}{1}\right)^{-2}$ (1); $= \left(\frac{1}{4}\right)^2$ (1); value $\frac{1}{16}$ (1).	1+1+1
17	(i) 4.5×10^7 (1); (ii) 7.8×10^{-7} (1); (iii) 2.3×10^3 (1).	1+1+1
	OR: (i) 4^{-2} (1); (ii) 4^{-3} (1); (iii) 4^6 (1).	1+1+1
18	Add the exponents: $\left(\frac{3}{5}\right)^{-2}$ (1); $= \left(\frac{5}{3}\right)^2$ (1); value $\frac{25}{9}$ (1).	1+1+1
	OR: (i) 1.86×10^5 (1); (ii) 3.1×10^{-6} (1); (iii) 7.2×10^7 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $x + 5 = 13$, $x = 8$ (2); (ii) $6x = 18$, $x = 3$ (2); (iii) $x - 6 = -4$, $x = 2$ (1).	2+2+1
	OR: $72000 = 7.2 \times 10^4$ (1); $0.012 = 1.2 \times 10^{-2}$ (1); product 8.64×10^2 (2); quotient 6×10^6 (1).	1+1+2+1
20	(i) $x + 2 = 7$, $x = 5$ (2); (ii) $4x = 28$, $x = 7$ (2); (iii) $x - 2 = -5$, $x = -3$ (1).	2+2+1
	OR: $0.0005 = 5 \times 10^{-4}$ (1); $0.000002 = 2 \times 10^{-6}$ (1); product 1×10^{-9} (2); quotient 2.5×10^2 (1).	1+1+2+1

Mistakes to avoid:

- Writing $2^{-3} = -8$ or $-\frac{1}{8}$; a negative exponent means the **reciprocal**, so $2^{-3} = \frac{1}{8}$.
- Adding the bases: $2^3 \times 2^4 = 4^7$ instead of $2^{3+4} = 2^7$.
- Saying $a^0 = 0$; for $a \neq 0$ it is 1.
- Multiplying exponents when multiplying powers of the same base ($a^m \times a^n = a^{mn}$) or adding them in $(a^m)^n$.
- Leaving the answer as 25×10^4 or 0.25×10^6; in standard form the number part must satisfy $1 \leq m < 10$.
- Getting the sign of the exponent wrong for small numbers: $0.0003 = 3 \times 10^{-4}$, not 3×10^4.

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